Allyson Costa

Boulder, CO | 303-501-3845 | allyjcosta@gmail.com | linkedin.com/in/allyson-costa11 | allyjcosta.github.io/site/

Education

University of Colorado Boulder — Boulder, CO
Bachelor of Science in Electrical Engineering | GPA: 3.629 — *Current*

Technical Skills

Programming: Python, MATLAB, C, C++, Verilog HDL
Python Tools: NumPy, SciPy, Matplotlib, serial communication, FFT and spectral analysis
Embedded Systems: STM32F429, Arduino Uno, GPIO, I^2C, SPI, microcontroller programming
Circuit Design & Simulation: LTspice, SIMetrix, analog circuit design, filtering, feedback control
Test & Measurement: Oscilloscopes, Digilent Analog Discovery 3, WaveForms, data acquisition
Licensing: FCC Amateur Radio License

Experience

Funded Research Assistant- Characterization of Capacitor Nonlinearity — Fall 2026 – Present
University of Colorado Boulder — *Boulder, CO*

- Developing an automated measurement system to characterize multilayer ceramic capacitor nonlinearity using capacitance-versus-voltage measurements.
- Characterizing dynamic nonlinear behavior using FFT-based harmonic distortion analysis and automated data acquisition.

Funded Research Assistant- Spectrometer for Noise Measurement — Summer 2026
University of Colorado Boulder — *Boulder, CO*

- Designed and developed a low-cost software-defined noise spectrometer capable of measuring electronic component noise from 0.1 Hz to 10 kHz.
- Developed Python and microcontroller software for data acquisition, FFT-based spectral analysis, and visualization of frequency-domain data.
- Designed and tested low-noise amplification, filtering, and AC-coupling circuitry and characterized the system noise floor.
- Measured Johnson thermal noise from 1 kΩ to 1 MΩ resistors and validated experimental results within a 10-week research program.

Manager & Barista — Summer 2020 – 2026
East Simpson Coffee Company — *Lafayette, CO*

- Managed approximately 15 employees, including scheduling, inventory, merchandise ordering, and daily shop operations.

Projects

NASA Space Grant High-Altitude Balloon Payload | *Project Lead* — Fall 2024 – Summer 2025

- Led the design and launch of a high-altitude balloon payload to approximately 30 km in collaboration with Edge of Space Sciences.
- Integrated sensors and live telemetry for radiation, IMU, humidity, temperature, and pressure measurements.
- Coordinated team responsibilities, budget, system requirements, and testing to maintain data transmission and payload functionality during flight.

Closed-Loop Robot Motor Control System | *Analog Control, PWM, SPICE* — Spring 2026

- Designed and implemented a closed-loop motor control system using encoder feedback, an I-compensator, PWM generation, and motor-driver circuitry.
- Built and debugged the system through hands-on laboratory testing and SPICE circuit simulation.

STM32 Embedded Systems | *C, STM32F429* — Fall 2026 – Present

- Develop embedded C applications using direct memory addressing, register-level GPIO control, polling, and hardware peripherals on an STM32F429.

FPGA Digital Synthesizer | *Verilog HDL, FPGA* — Spring 2026

- Developed a digital synthesizer in Verilog HDL that generated audio frequencies based on user input.
- Implemented digital logic for square-wave audio generation and PWM-based volume control.